

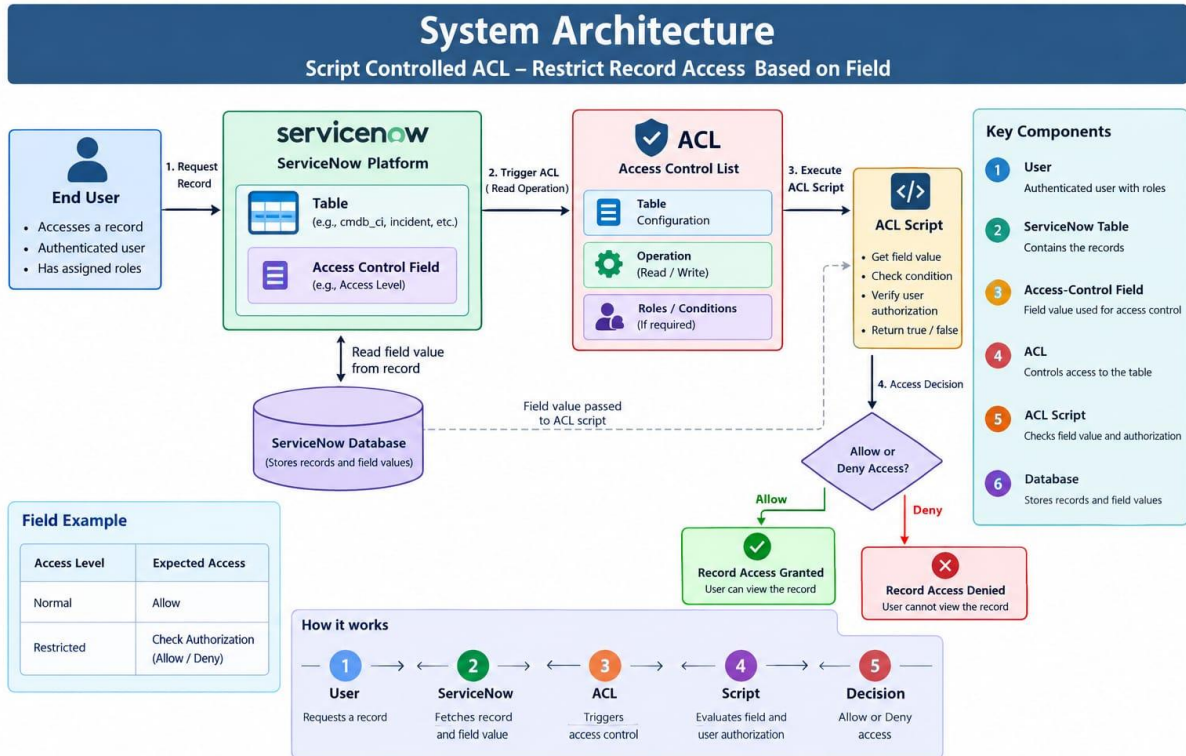
Script-Controlled ACL – Restrict Record Access Based on Field Value

Team Leader	A.Parimala ,asaluau020224128026
Team members	S.Priyanka ,asaluau020224128028
	R.Rathika ,asaluau020224128030
	M.Rajeswari ,asaluau020224128029
	S.Mounika ,asaluau020224128025

1. Objectives

This lab demonstrates how to implement a **script-controlled Access Control List (ACL)** to restrict users from viewing records unless they meet specific conditions. In this scenario, only users belonging to the **EEE branch** are allowed to view records where the **Branch field is set to EEE**, while administrators retain full access.

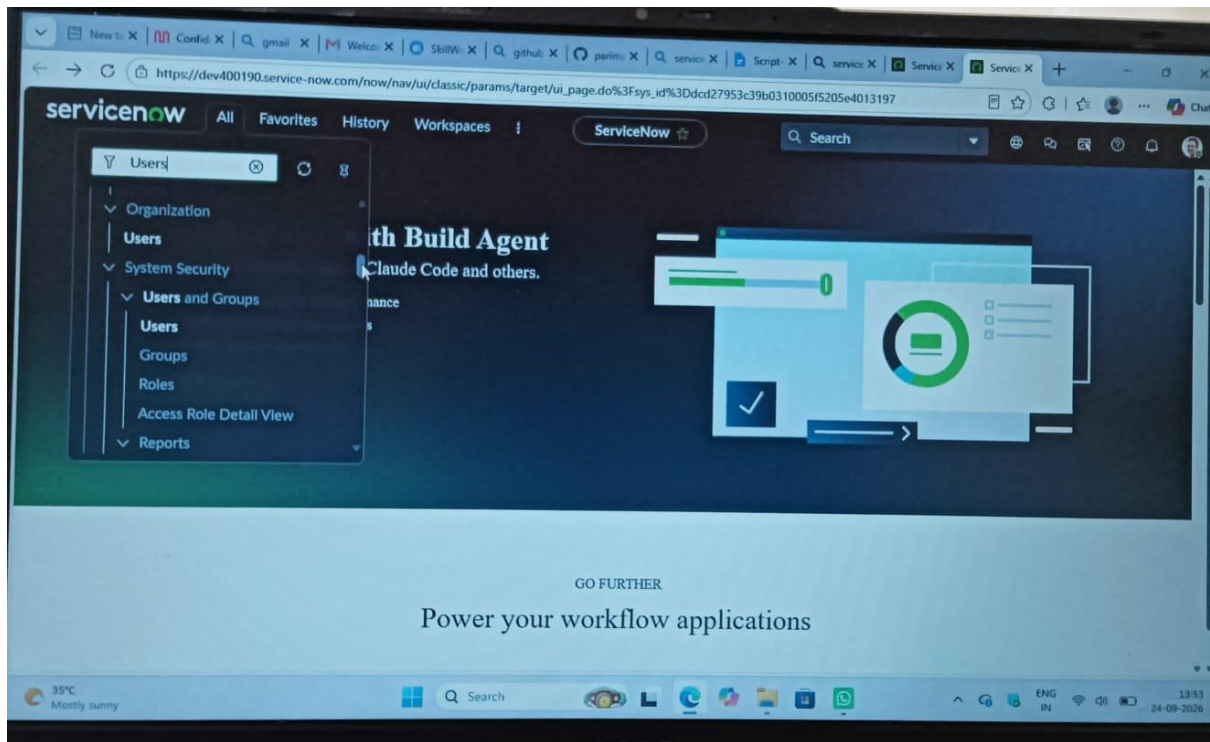
2.System Architecture:



Milestone-1 Creation of Users and Roles

User Creation:

- Log in** to the ServiceNow instance with **admin access**.
- Create a test user:**
 - Navigate to **User Administration** → **Users**
 - Click **New**
 - User ID:** EEE User
 - First name:** EEE
 - Last name:** User
 - Email: eeuser@gmail.com
 - Save the record



ServiceNow Developers | New Record | User | ServiceNow | Ask Gemini

dev400190.service-now.com/now/nav/ui/classic/params/target/sys_user.do%3Fsys_id%3D-1%26sys_is_list%3Dtrue%26sys_target%3Dsys_user%26sysparm_checked_ite...

ServiceNow | All | Favorites | History | Workspaces | User - New Record | Search

User - New record | Submit

To set up the User's password, save the record and then click Set Password.

User ID	EEE User	Email	eeeuser@gmail.com
First name	EEE	Language	-- None --
Last name	User	Calendar integration	Outlook
Title		Time zone	System (America/Los_Angeles)
Department		Date format	System (2026-08-19)
Password needs reset	☐	Business phone	
Locked out	☐	Mobile phone	
Active	☒	Photo	Click to add...
Identity type	Human		
Internal Integration User	☐		

Submit

Related Links

3.Role Creation:

Navigate to **User Administration → Roles**

- Click **New**
- **Name:** Enter role name as bb1
- **Description:** Brief role purpose
- Click **Submit**
- Create other roles also like **bb2,bb3,bb4.(follow the same procedure)**
- Assign all the roles to the EEE User

The screenshot shows the ServiceNow 'Role - New Record' form. The form is titled 'Role - New Record' and has a 'Submit' button at the bottom left. The form fields are:

- Name:** bb 1
- Application:** Global
- Elevated privilege:** ☐
- Description:** (empty text area)

4.Tables Creation:

1. In the **Application Navigator**, search for **System Definition → Tables**.

2. Click **New** and create a custom table with the following details:

- **Label:** Institution Details
- **Name:** u_institution_details
- **Extends for :** false

The screenshot shows the ServiceNow 'Table - New Record' form. The form is titled 'Table - New Record' and has a search bar at the top. Below the search bar, there is a yellow banner with a message: 'ServiceNow recommends creating custom tables in scoped applications. To learn more about creating scoped applications, click [here](#).' Below this, there is a blue box with a description: 'A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)'.

The form contains the following fields and controls:

- Label:** Institution Details
- Name:** u_institution_details
- Extends table:** (empty field with a search icon)
- Application:** Global
- Create module:** ☒
- Create mobile module:** ☒
- Add module to menu:** -- Create new --
- New menu name:** Institution Details
- Remote Table:** ☐

At the bottom of the form, there are tabs for 'Columns', 'Controls', and 'Application Access'. The 'Columns' tab is currently selected, showing a 'Table Columns' section with a search bar and a 'Dictionary Entries' section.

3. Click **Submit**.

4. Add the following fields to the **u_institution_details** table:

- **Student Roll Number**– Auto Number
- **Student Name** – Reference- User
- **Faculty Name** – Reference- User
- **Branch** – Choice (ECE, EEE, CSE)
- **Email**– String
- **Phone Number**– String
- **Description**– Multi String

5. Save the Table: Create multiple records in the **Student Records** table with different branch values (ECE, EEE, CSE).

6. Create multiple records in the **Student Records** table with different branch values (ECE, EEE, CSE).

The screenshot shows the ServiceNow interface for the 'Table - Institution Details'. The form contains the following fields:

Field	Type	Value	Length	Required
Created	Date/Time	(empty)	40	false
Updated	Date/Time	(empty)	40	false
Updates	Integer	(empty)	40	false
Sys ID	Sys ID (GUID)	(empty)	32	false
Created by	String	(empty)	40	false
Student Roll Number	String			false
Student Name	Reference	User		false
Faculty Name	Reference	User		false
Branch	Choice			false
Email	String			false
Phone Number	String			false
Description	String			false

Buttons: Delete, Update, Delete All Records

Related Links: Form Builder, Design Form

5. Creation of Access Control Lists(Read ACL):

1. In the **Application Navigator**, search for **System Security** → **Access Control (ACL)**.
2. Elevate “Security_admin” role
3. Click **New** to create a new ACL.
4. Enter the following details:
 - **Type:** record
 - **Operation:** read
 - **Name:** u_institution_details
 - **Active:** true

- **Advanced:** true

5. In the **Requires role** related list, add:

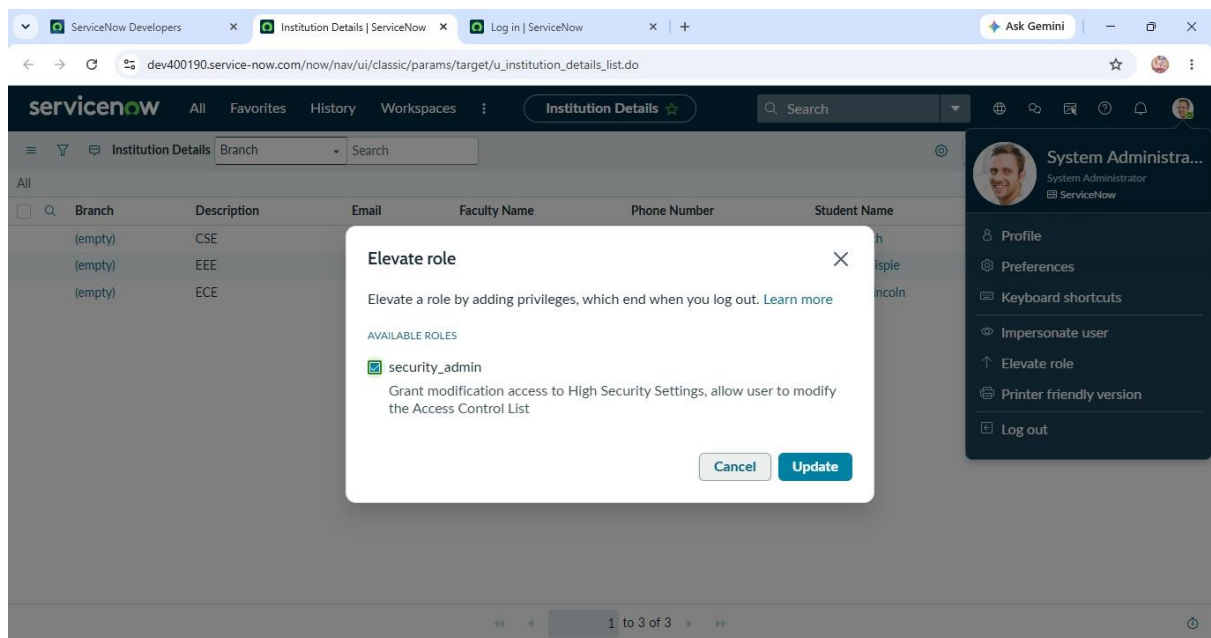
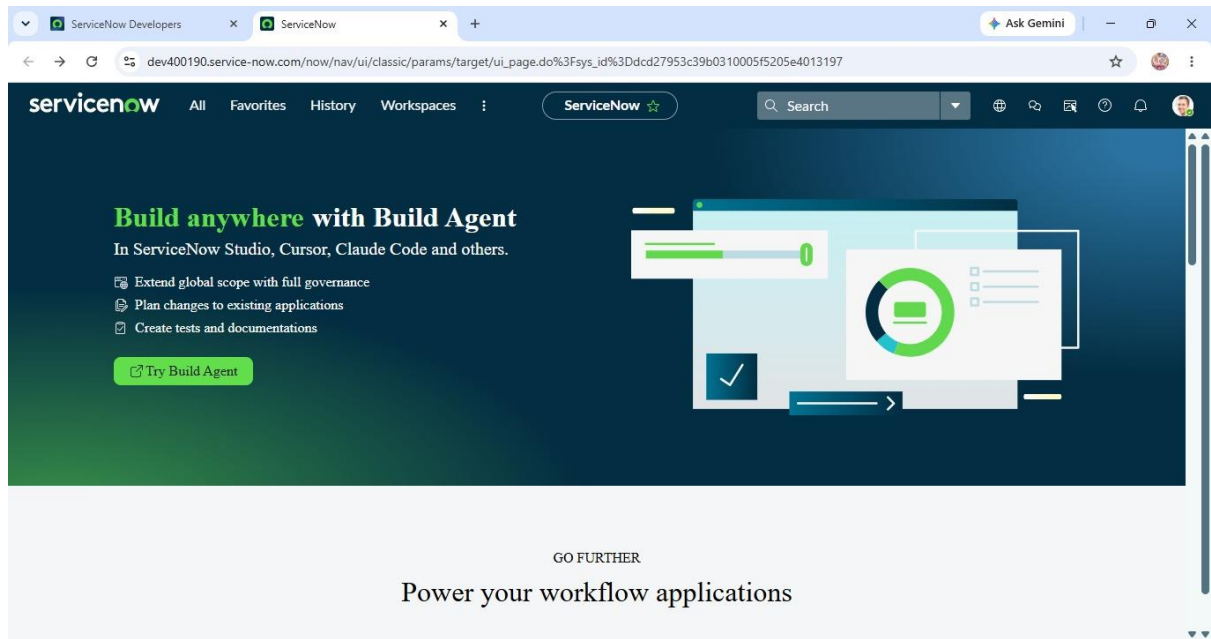
- bb1 (custom role)

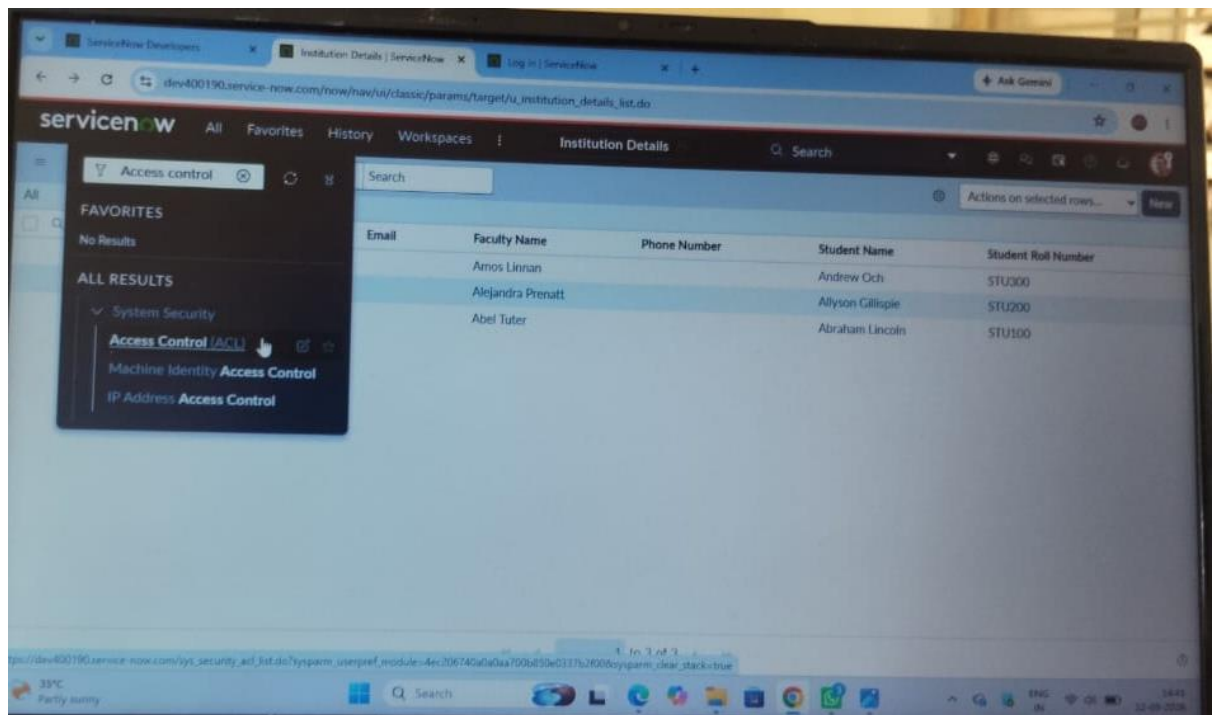
6. Add Data condition as **Branch is EEE**

7. In the **Script** field, add the following script:

```
(function () {  
  
  // Allow admin users full access  
  
  if (gs.hasRole('admin')) {  
  
    return true;  
  
  }  
  
  // Allow only EEE branch users to see EEE records  
  
  if (gs.hasRole('bb1')){  
  
    return true;  
  
  }  
  
  // Deny access for all others  
  
  return false;  
  
})();
```

8. Click **Submit** to save the ACL.





The screenshot shows the ServiceNow web interface for "Access Controls". The table lists various access control records with the following columns: Name, Decision Type, Operation, Type, Active, Updated by, and Updated.

Name	Decision Type	Operation	Type	Active	Updated by	Updated
u_institution_details	Allow If	write	record	true	admin	2026-09-22 00:52:54
u_institution_details	Allow If	delete	record	true	admin	2026-09-22 00:52:54
u_institution_details	Allow If	create	record	true	admin	2026-09-22 00:52:53
u_institution_details	Allow If	read	record	true	admin	2026-09-22 00:52:53
u_institution_details	Allow If	read	record	true	admin	2026-09-22 02:18:53
u_licensing_exclusion	Allow If	create	record	true	admin	2017-08-15 13:09:53
u_licensing_exclusion	Allow If	delete	record	true	admin	2017-08-15 13:10:11
VA Channel Integration ACL	Allow If	execute	REST_Endpoint	true	admin	2023-04-05 00:28:13
VA Designer Config	Allow If	execute	REST_Endpoint	true	admin	2024-05-22 12:09:23
VaCallbackPropertyUtil	Allow If	execute	client_callable_script_include	true	admin	2022-05-09 01:51:24
ValidateAesVersion	Allow If	execute	client_callable_script_include	true	admin	2022-01-11 10:48:04
Valid employee profile user	Allow If	execute	REST_Endpoint	true	admin	2024-06-12 13:17:06
ValidateAlternatePortal	Allow If	execute	client_callable_script_include	true	admin	2024-03-06 16:53:36

The table is paginated, showing 1 to 20 of 762 records.

ServiceNow Developers | Access Controls | ServiceNow | Log in | ServiceNow | Ask Gemini

dev400190.service-now.com/now/nav/ui/classic/params/target/sys_security_acl_list.do%3Fsysparm_first_row%3D1%26sysparm_query%3DGOTOname%253e%253du_instituti...

servicenow All Favorites History Workspaces Access Controls Search

Access Control u_institution_details Update Delete

* Type: record ⓘ

* Operation: read ⓘ

Admin overrides: ☒

Decision Type: Allow If

Protection policy: -- None --

* Name: Institution Details [u_institution_details]

Description: Allow read for records in u_institution_details, for users with role bb 1, and if the ACL script returns true, and if the ACL condition (u_branch=EEE^EQ) evaluates to true.

Application: Global ⓘ

Active: ☒

Advanced: ☒

Conditions

Requires role

Role
bb 1

ServiceNow Developers | u_institution_details | Access Co... | Log in | ServiceNow | Ask Gemini

dev400190.service-now.com/now/nav/ui/classic/params/target/sys_security_acl.do%3Fsys_id%3Db0112168c32fc310005f5205e40131ce%26sysparm_record_target%3Dsys_se...

servicenow All Favorites History Access Control - u_institution_details Search

Access Control u_institution_details Update Delete

Data Condition

Condition: No. of records matching the condition: 0 ⓘ

Add Filter Condition Add OR Clause

Branch is EEE AND OR X

Advanced Condition

Controlled by References ⓘ

Script: ☒ Turn on ECMAScript 2021 (ES12) mode ⓘ

```

1 function () {
2   // Allow admin users full access
3   if (gs.hasRole('admin')) {
4     return true;
5   }
6   // Allow only EEE branch users to see EEE records
7   if (gs.hasRole('bb1')) {
8     return true;
9   }
10  // Deny access for all others
11  return false;

```

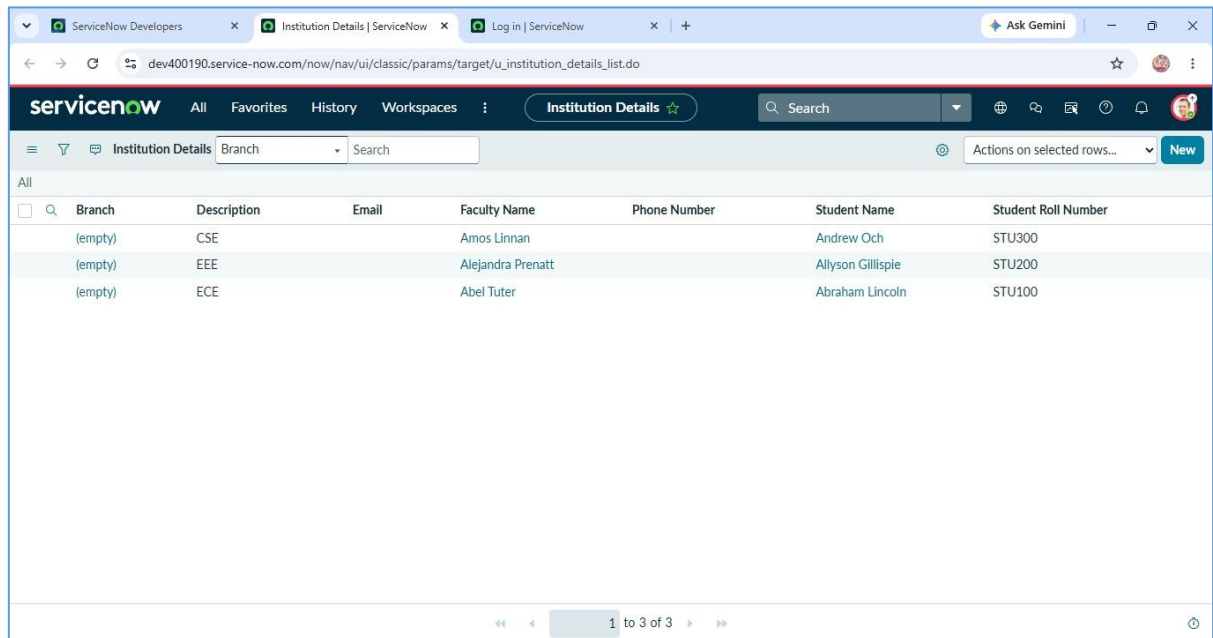
9.Impersonate **EEE users** and open the Student Records list.

10.Copy the table name u_institution_details.list/LIST and paste in the All menu

11. Impersonate **Admin** and open the Student Records list.

Verify :

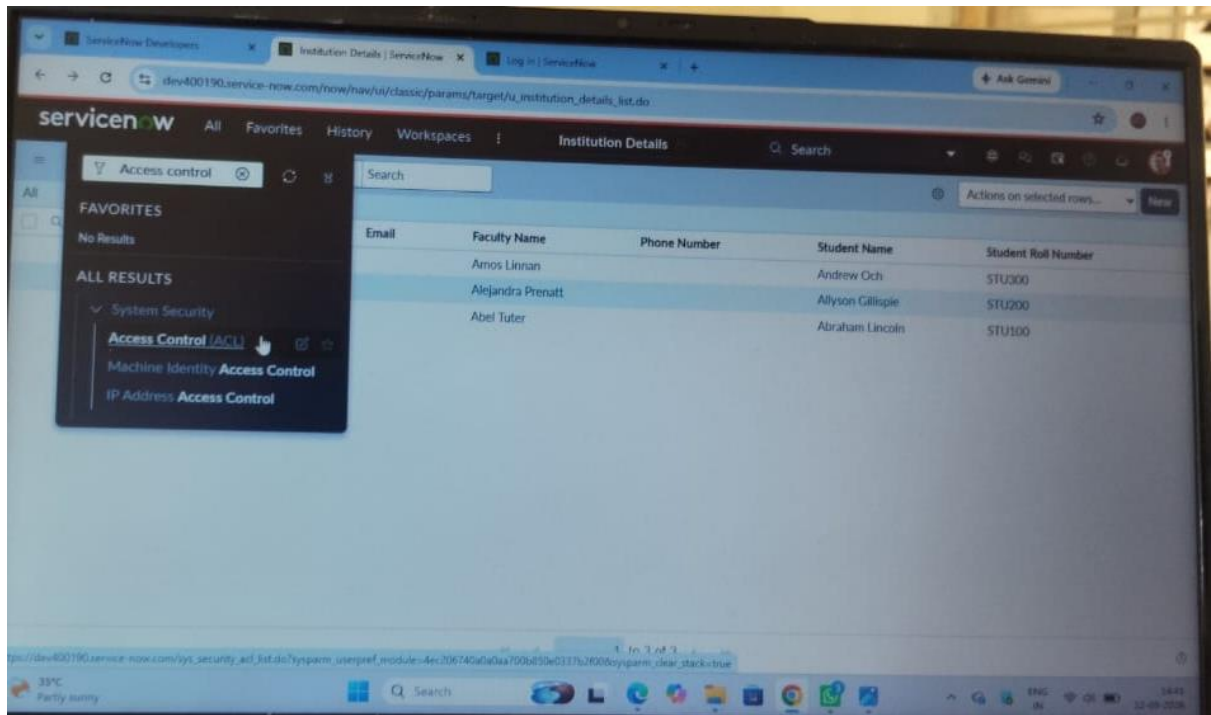
- User with **bb1** role can view **only** **EEE** branch records

A screenshot of a web browser displaying the ServiceNow 'Institution Details' list. The browser has three tabs: 'ServiceNow Developers', 'Institution Details | ServiceNow', and 'Log in | ServiceNow'. The address bar shows the URL 'dev400190.service-now.com/now/nav/ui/classic/params/target/u_institution_details_list.do'. The ServiceNow interface includes a top navigation bar with 'All', 'Favorites', 'History', and 'Workspaces'. Below this is a search bar and a 'New' button. The main content area shows a table with columns: Branch, Description, Email, Faculty Name, Phone Number, Student Name, and Student Roll Number. The table contains three rows of data. The first row has an empty branch, CSE description, and Andrew Och student. The second row has an empty branch, EEE description, and Allyson Gillispie student. The third row has an empty branch, ECE description, and Abraham Lincoln student. The bottom of the screen shows a pagination bar indicating '1 to 3 of 3' records.

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
(empty)	CSE		Amos Linnan		Andrew Och	STU300
(empty)	EEE		Alejandra Prenatt		Allyson Gillispie	STU200
(empty)	ECE		Abel Tuter		Abraham Lincoln	STU100

7. Creation of Access Control Lists(Create ACL):

1. Click **New** to create a new ACL(Create)
2. Enter the following details:
 - **Type:** record
 - **Operation:** create
 - **Name:** u_institution_details
 - **Active:** true
3. In the **Requires role** related list, add:
 - bb2 (custom role)
4. No need to add any Data condition
5. Save the ACL Record



Access Controls							
All > Name >= u_institution_details							
☐	Name	Decision Type	Operation	Type	Active	Updated by	Updated
☐	Search	Search	Search	Search	Search	Search	Search
☐	u_institution_details	Allow If	write	record	true	admin	2026-09-22 00:52:54
☐	u_institution_details	Allow If	delete	record	true	admin	2026-09-22 00:52:54
☐	u_institution_details	Allow If	create	record	true	admin	2026-09-22 00:52:53
☐	u_institution_details	Allow If	read	record	true	admin	2026-09-22 00:52:53
☐	u_institution_details	Allow If	read	record	true	admin	2026-09-22 02:18:53
☐	u_licensing_exclusion	Allow If	create	record	true	admin	2017-08-15 13:09:53
☐	u_licensing_exclusion	Allow If	delete	record	true	admin	2017-08-15 13:10:11
☐	VA Channel Integration ACL	Allow If	execute	REST_Endpoint	true	admin	2023-04-05 00:28:13
☐	VA Designer Config	Allow If	execute	REST_Endpoint	true	admin	2024-05-22 12:09:23
☐	VaCallbackPropertyUtil	Allow If	execute	client_callable_script_include	true	admin	2022-05-09 01:51:24
☐	ValidateAesVersion	Allow If	execute	client_callable_script_include	true	admin	2022-01-11 10:48:04
☐	Valid employee profile user	Allow If	execute	REST_Endpoint	true	admin	2024-06-12 13:17:06
☐	ValidateAlternatePortal	Allow If	execute	client_callable_script_include	true	admin	2024-03-06 16:53:36

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type: record
 * Operation: create
 Admin overrides: ☒
 Decision Type: Allow If
 Protection policy: -- None --
 * Name: Institution Details [u_institution_details]
 Description:
 Application: Global
 Active: ☒
 Advanced: ☐

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Role: bb 2

Security Attribute Condition

Local or Existing: ☒ Existing ☐ Local

Security Attribute:
 Condition: All of these conditions must be met
 -- choose field --

Verify :

- User with **bb1 &bb2** roles can view **only EEE branch records b New** Button

8.Creation of Access Control Lists(Write ACL):

1. Click New to create a new ACL(Write)

2. Enter the following details:

- Type: record
- Operation: Write
- Active: true

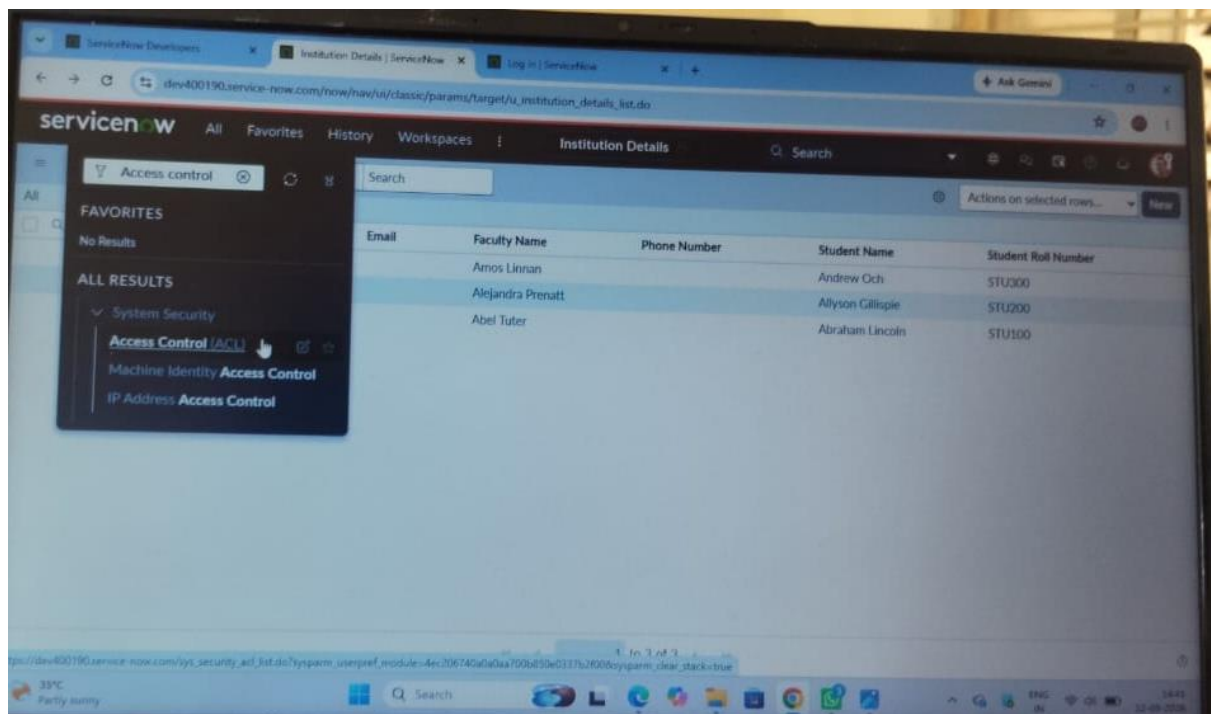
1. In the Requires role related list, add:

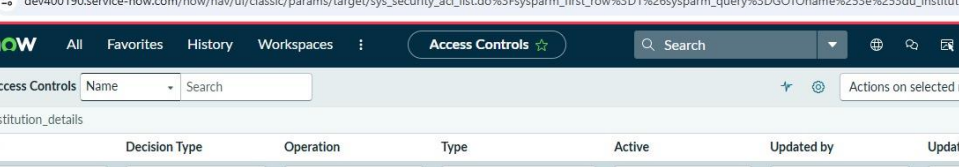
- bb3 (custom role)

2. No need to add any Data condition

3. Save the ACL Record

- Name: u_institution_details





The screenshot displays the ServiceNow Access Controls interface. At the top, there are navigation tabs for 'ServiceNow Developers', 'Access Controls | ServiceNow', and 'Log in | ServiceNow'. Below the tabs, the breadcrumb path is 'dev400190.servicenow.com/now/nav/ui/classic/params/target/sys_security_acl_list.do?3Fsyparm_first_row%3D1%26syparm_query%3DGOTOName%253e%253du_instituti...'. The main header shows 'servicenow' with tabs for 'All', 'Favorites', 'History', and 'Workspaces'. The 'Access Controls' tab is active, and a search bar is present. Below the header, a table lists access rules. The table has columns: Name, Decision Type, Operation, Type, Active, Updated by, and Updated. The first row is 'u_institution_details' with 'Allow If' decision type, 'write' operation, 'record' type, and is active. The second row is 'u_institution_details' with 'Allow If' decision type, 'delete' operation, 'record' type, and is active. The third row is 'u_institution_details' with 'Allow If' decision type, 'create' operation, 'record' type, and is active. The fourth row is 'u_institution_details' with 'Allow If' decision type, 'read' operation, 'record' type, and is active. The fifth row is 'u_institution_details' with 'Allow If' decision type, 'read' operation, 'record' type, and is active. The sixth row is 'u_licensing_exclusion' with 'Allow If' decision type, 'create' operation, 'record' type, and is active. The seventh row is 'u_licensing_exclusion' with 'Allow If' decision type, 'delete' operation, 'record' type, and is active. The eighth row is 'VA Channel Integration ACL' with 'Allow If' decision type, 'execute' operation, 'REST_Endpoint' type, and is active. The ninth row is 'VA Designer Config' with 'Allow If' decision type, 'execute' operation, 'REST_Endpoint' type, and is active. The tenth row is 'VaCallbackPropertyUtil' with 'Allow If' decision type, 'execute' operation, 'client_callable_script_include' type, and is active. The eleventh row is 'ValidateAesVersion' with 'Allow If' decision type, 'execute' operation, 'client_callable_script_include' type, and is active. The twelfth row is 'Valid employee profile user' with 'Allow If' decision type, 'execute' operation, 'REST_Endpoint' type, and is active. The thirteenth row is 'ValidateAlternatePortal' with 'Allow If' decision type, 'execute' operation, 'client_callable_script_include' type, and is active. The bottom of the table shows '1 to 20 of 762' items.

Name	Decision Type	Operation	Type	Active	Updated by	Updated
u_institution_details	Allow If	write	record	true	admin	2026-09-22 00:52:54
u_institution_details	Allow If	delete	record	true	admin	2026-09-22 00:52:54
u_institution_details	Allow If	create	record	true	admin	2026-09-22 00:52:53
u_institution_details	Allow If	read	record	true	admin	2026-09-22 00:52:53
u_institution_details	Allow If	read	record	true	admin	2026-09-22 02:18:53
u_licensing_exclusion	Allow If	create	record	true	admin	2017-08-15 13:09:53
u_licensing_exclusion	Allow If	delete	record	true	admin	2017-08-15 13:10:11
VA Channel Integration ACL	Allow If	execute	REST_Endpoint	true	admin	2023-04-05 00:28:13
VA Designer Config	Allow If	execute	REST_Endpoint	true	admin	2024-05-22 12:09:23
VaCallbackPropertyUtil	Allow If	execute	client_callable_script_include	true	admin	2022-05-09 01:51:24
ValidateAesVersion	Allow If	execute	client_callable_script_include	true	admin	2022-01-11 10:48:04
Valid employee profile user	Allow If	execute	REST_Endpoint	true	admin	2024-06-12 13:17:06
ValidateAlternatePortal	Allow If	execute	client_callable_script_include	true	admin	2024-03-06 16:53:36

ServiceNow Developers | New Record | Access Control | Log in | ServiceNow | Ask Gemini

dev400190.service-now.com/now/nav/ui/classic/params/target/sys_security_acl.do%3Fsys_id%3D-1%26sys_is_list%3Dtrue%26sys_target%3Dsys_security_acl%26sysparm_che...

servicenow | All | Favorites | History | Admin | Access Control - New Record | Search

Access Control New record | Submit

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Role
bb3

+ Insert a new row...

Security Attribute Condition

Local or Existing ☒ Existing ☐ Local

* Security Attribute

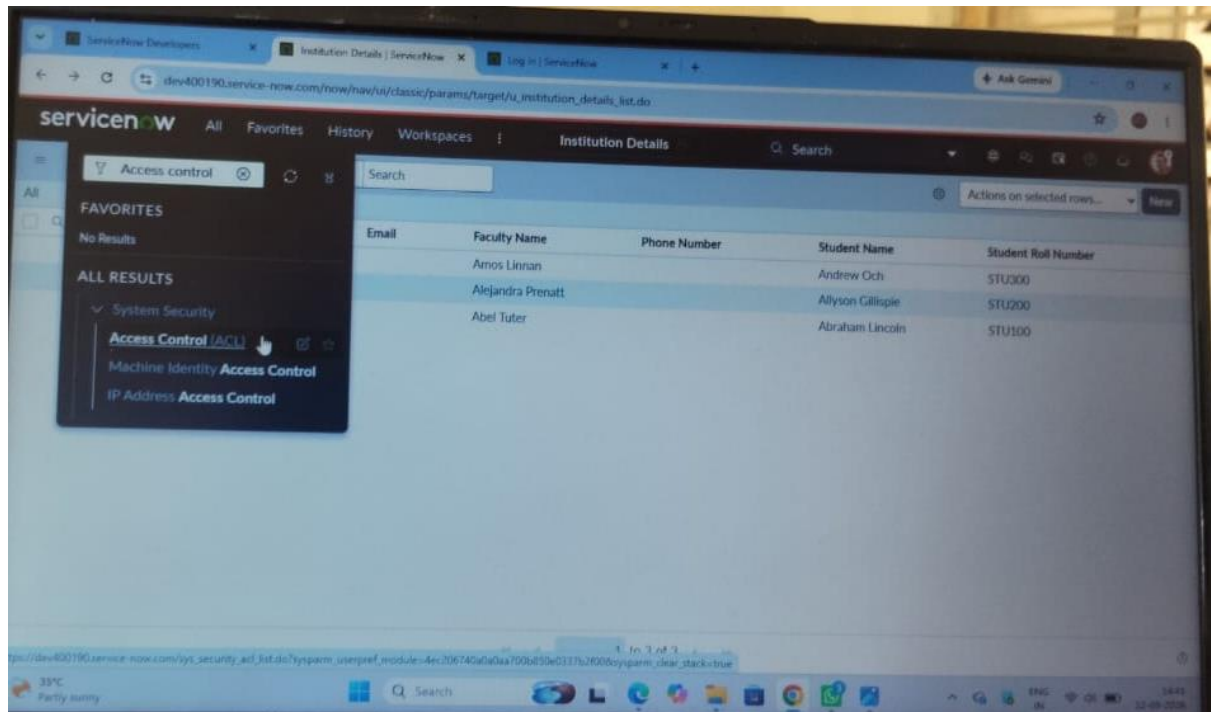
Verify :

- User with **bb1 &bb2 &bb3** roles can view and edit the **EEE branch records along with New Button**

9.Creation of Access Control Lists(Delete ACL):

1. Click **New** to create a new ACL(Delete)
2. Enter the following details:
 - **Type:** record
 - **Operation:** Delete
 - **Name:** u_institution_details
 - **Active:** true
3. In the **Requires role** related list, add:
 - bb4 (custom role)
4. No need to add any Data condition

5. Save the ACL Record



The screenshot shows the 'Access Control - New Record' form in ServiceNow. The form includes the following fields and options:

- * Type: record
- * Operation: delete
- Application: Global
- Active: ☒
- Admin overrides: ☒
- Decision Type: Allow If
- Protection policy: -- None --
- * Name: Institution Details [u_institution_details]
- Description: (empty text area)

A warning message is displayed at the top: "Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL."

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Role

bb4

Security Attribute Condition

Local or Existing ☒ Existing ☐ Local

Security Attribute

Condition All of these conditions must be met

-- choose field --

OR AND

New Criteria

Verify :

- Users with **bb1 & bb2 & bb3 & bb4** roles can see, create and edit along with delete the **EEE branch records**.

10.Outcomes:

By completing this lab, learners gain a comprehensive understanding of how script-controlled READ, WRITE, CREATE, and DELETE ACLs enforce record-level security, how user roles and record field values are evaluated together to make access decisions, and how ACLs protect sensitive data from unauthorized viewing, modification, creation, and deletion across forms, lists, and Service Portal interfaces.